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MILITARY SPECIFICATION

LOADING JACK SYSTEM, ISO CONTAINER

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance, design, and test requirements for the ISO container loading jack system (ILJS). The ILJS shall provide level loading and unloading on C-130 and C-141 aircraft, on and off loading of 102-inch wide flat bed trailers and all 25K-loaders, and mobility and maneuverability for freight containers having ISO corner fittings (hereinafter referred to as ISO containers).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-B-601 - Boxes, Wood, Cleated-Plywood.

|Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.|

AMSC N/A

FSC 3950
FSC 8145

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MIL-P-116 - Preservation, Methods of.
MIL-S-8512 - Support Equipment, Aeronautical, Special, General Specification for the Design of.
MIL-P-15024/6 - Plates, Identification, Equipment.
MIL-B-26195 - Boxes, Wood-Cleated, Skidded, Load-Bearing Base.
MIL-P-27443 - Pallets, Cargo, Aircraft, Type HCU-6/E, HCU-12/E, and HCU-10/E.
MIL-C-46168 - Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-P-53022 - Primer, Epoxy Coating, Corrosion Inhibiting, Lead and Chromate Free.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-210 - Climatic Extremes for Military Equipment.
MIL-STD-642 - Identification Marking of Combat and Tactical Transport Vehicles.
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines.
MIL-STD-1366 - Material Transportation System Dimensional and Weight Constraints, Definition.
MIL-STD-1472 - Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1791 - Designing for Internal Aerial Delivery in Fixed Wing Aircraft.
MS51336 - Lunette-Coupler, Drawbar, Ring.

FEDERAL

FED-STD-595 - Colors.

2.1.2 Other Government documents. The following other Government documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

AIR FORCE SYSTEMS COMMAND DESIGN HANDBOOK (DH)

DH 1-11 - Air Transportability.

(Copies of other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS shall be the issue of the non-Government documents which is current on the date of the solicitation.

AMERICAN WELDING SOCIETY, INC. (AWS)

D1.1 - Structural Welding Code - Steel.

(Application for copies should be addressed to the American Welding Society, Inc., 550 N.W. LeJeune Road, P.O. Box 351040, Miami, FL 33135.)

INTERNATIONAL ORGANIZATION OF STANDARDIZATION (ISO)

ISO 1161 - Series 1 Freight Containers - Corner Fittings.

(Application for copies should be addresssed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

STEEL STRUCTURES PAINTING COUNCIL (SSPC)

SSPC-SP10 - Near White Blast Cleaning.

(Application for copies should be addressed to the Steel Structures Painting Council, 4400 Fifth Avenue, Pittsburgh, PA 15213.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Description. The ILJS shall consist of four corner jack assemblies attachable to the top and bottom ISO corner fittings of 4'3" X 8 X 20 and 8 X 8 X 20 ISO containers. The ILJS shall provide level loading and unloading of containers on C-130 and C-141 aircraft, 102-inch wide flat bed trailers, and all 25K-loaders. The ILJS shall include a towing assembly which will allow mobility and maneuverability for loading/off loading and positioning the ILJS/container combination. The ILJS shall be designed to meet all requirements as defined herein.

3.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.2.1 and 6.2). Examination and tests shall be as specified in section 4; and, all examinations and tests shall be conducted by the contractor subject to surveillance and approval by the Government (see 6.4). The Government may witness and/or perform in any or all of the first article examination and tests, in addition to the contractor's first article examination and tests.

3.3 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.4 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.5 ILJS structure and components. The ILJS structure and all of its components shall be designed to a minimum safety factor of 2.5 based on the material yield stress, applied to the ILJS rated load of 25,000 pounds (lb).

3.5.1 Attachment points. The ILJS attachment points shall be restricted to the container side and end wall apertures of the ISO corner fittings (ISO 1161), located at the top and bottom of the container. The ILJS shall be capable of being attached and removed while the four bottom ISO corners of the container are sitting on the ground.

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3.6 Corner jack assemblies. Four corner jack assemblies shall be provided. Each corner jack assembly shall connect to the ISO container at the top and bottom corner fittings. Each corner jack assembly shall be capable of being attached to any of the top and bottom corner fittings. Each corner jack assembly shall be mounted on a two-wheel pneumatic-tired caster assembly. When tested in accordance with MIL-S-8512, each corner jack shall withstand a static load of 20,000 lb without showing signs of deterioration or permanent deformation. The jacks shall be marked at one-inch increments; each sixth increment shall be highlighted and numbered.

3.6.1 Assembly components. All assembly components of the ILJS shall be identical to like components to insure interchangeability of component parts of the ILJS. Assembly components shall meet the weight requirements specified in 3.8.2.

3.6.2 Caster assembly. Each caster assembly shall caster in 360 degree rotation in any direction (fully articulating). The caster assembly shall not have any side to side tilting capability. A mechanism shall be provided to lock the caster assembly at 90-degree positions in line with container side walls and ends.

3.6.3 Tires. The type and size of tires furnished shall have adequate capability for the ILJS operation as specified in this specification when loaded in accordance with 3.9. The tires shall be pneumatic.

3.6.4 Brakes. Each corner jack wheel assembly shall be provided with a single hand operated brake that will have the capability to perform as both a parking brake, when fully engaged, and have a partial (dynamic) braking action as an interim operation which would allow partial braking in any intermediate position. Brake actuation forces shall not exceed 270 inch-pounds (15 lb on an 18-inch handle). The brake assembly shall hold the fully loaded ILJS on a 7 percent slope. Brake devices consisting of shoes bearing on the tire or ground are not acceptable.

3.7 Towing assembly. The towing assembly shall enable the ILJS to be positively steered and towed in the forward direction; and positively steered and pushed in the reverse (backing up) direction. The towing assembly shall be attached to the corner jack assemblies when they are either in the inboard or outboard configurations. The towing assembly shall have a lunette eye conforming to MS51336 and shall attach to any tow vehicle with a pintle hook height of 10 to 40 inches from the ground. The towing assembly shall attach to either end of the container. The towing assembly shall be attachable/detachable in any operational configuration of the ILJS.

3.7.1 Caster swiveling bars. Four caster swiveling bars shall be provided. Each bar shall enable an individual caster assembly to be rotated. Each bar shall swivel any caster.

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3.8 Physical characteristics.

3.8.1 Dimensions and clearance. The ILJS, when atatched to an ISO container, shall conform to the following measurements:

a. The ILJS shall pick up the container off the ground (container not resting on skids or dunnage) and raise the container to a height of 66 inches as measured from ground surface to the bottom of the container.
b. The ILJS shall allow a minimum of a 10-1/2-foot space between the offset jacks during flatbed loading.
c. The ILJS shall be designed and configured to meet the requirements of 3.10.7.1. When straddling the aircraft loading ramp, the ILJS shall maintain a minimum 3-inch clearance between the jack legs and each side of the ramp.

3.8.2 ILJS weight. The ILJS shall be the minimum weight possible with a maximum weight of 500 lb per corner assembly and an overall system weight of 2,200 lb including tow bar and caster swiveling bars. No individual unit of the ILJS shall weigh more than 227 lb (see 3.16.1).

3.9 Payload. The ILJS shall transport a 4'3" X 8 X 20, or 8 X 8 X 20 ISO container with corner fittings conforming to ISO 1161. The payload (container and contents) shall not exceed 25,000 lb gross weight. The center of gravity (CG) of the container may vary longitudinally and laterally as shown in figure 1. The vertical CG shall not exceed 2.5 feet (ft) above the 4'3" X 8 X 20 container floor; 4.0 ft above the 8 X 8 X 20 container floor. The design of the ILJS shall be able to accommodate the variable CG. Any test payload shall be in a 20-foot ISO container weighing 25,000 lb with CG located within the cross-hatch area as shown in figure 1.

3.10 Performance. The following performance criteria shall be applicable to the ILJS configured in both the inboard and the outboard modes. Inboard mode is defined as the ILJS configured with the corner jack assemblies attached to the end walls of the ISO container. Outboard mode is defined as the ILJS configured with the corner jack assemblies attached to the side walls of the ISO container.

3.10.1 Lifting with ILJS. The ILJS shall be designed so that no mechanism shall require external power or externally renewed power other than the manpower provided by the crew to position, attach, lift, and prepare the container for movement. The container shall be lifted manually to 66 inches in not more than 15 minutes by four people.

3.10.2 Steering and turning ability. The ILJS, when fully loaded (see 3.9), raised to a towing height of 18 inches, and with prime mover operating, shall turn right and left without damage or interference with the towed container or prime mover. The 18-inch towing height shall be measured from ground surface to the bottom of the container.

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3.10.3 Controllability over surfaces. The ILJS, when at an 18-inch towing height and loaded in accordance with 3.9, shall exhibit stability when towed over surfaces specified in 3.10.10 and table I. The 18-inch towing height shall be measured from ground surface to the bottom of the container.

3.10.4 Longitudinal inclines. The ILJS, when at an 18-inch towing height and loaded in accordance with 3.9, shall follow the towing vehicle without weaving to an extent which would adversely affect the controllability of the ILJS when ascending or descending longitudinal inclines having a 7 percent slope.

3.10.5 Side slope. The ILJS, when at an 18-inch towing height and loaded in accordance with 3.9, shall follow straight behind the prime mover without slipping or upsetting when operating either left or right on a surface having a 5 percent side slope.

3.10.6 Backing of ILJS. The ILJS, when at an 18-inch towing height and loaded in accordance with 3.9, and towing assembly shall provide sufficient control to the container so that the prime mover can move the container backwards for a distance of 50 ft in a 10-foot wide lane.

3.10.7 Operational cycle. The operational cycle shall consist of the operations and time frames described herein.

3.10.7.1 Level loading. The ILJS, when loaded in accordance with 3.9 and in the outboard mode, shall be capable of level loading the ISO container on and off a C-130 and C-141 aircraft. At no time during level loading using the ILJS shall the container exceed aircraft floor loading limits in accordance with MIL-STD-1791. Level loading shall be accomplished in 20 minutes or less for each container. Loading shall commence when the fully loaded container (see 3.9), with pallets and loading jack attached, is raised to the required loading height and is backed up to within one foot of the aircraft cargo platform. Loading is complete when the container is in position to tiedown within the aircraft. Pallet dimensions shall conform to MIL-P-27443, type I (88 inches X 108 inches) which is commonly referred to as 463L pallet.

3.10.7.2 Flat bed truck and 25K-loader loading. The ILJS shall be configurable in the outboard position for loading ISO containers onto and off flat bed trucks and 25K-loaders. The ILJS shall be attachable from the ground to a truck-mounted or 25K-loader-mounted container. Loading shall be accomplished in 45 minutes or less. Loading shall be accomplished manually by no more than four personnel. Loading shall commence when the container is resting on the ground and jacks and all hardware are removed. Loading is complete when the container is resting on the flat bed truck or 25K-loader and all jacks and hardware are removed. Unloading shall be accomplished in 45 minutes or less. Unloading shall be accomplished manually by no more than four personnel. Unloading shall commence when the container is resting on the flat bed truck or 25K-loader and jacks and all hardware are removed. Unloading is complete when the container is resting on the ground and all jacks and hardware are removed.

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3.10.8 Wind load. The ILJS shall be capable of safely performing all functions in each operational cycle with a 30 mile per hour (mph), minimum, wind load on the 20-foot length of the container, when the jacks are at the 66-inch height and the wheels are restrained.

3.10.9 Parking brake performance. The parking brakes shall be sufficient to hold the container/ILJS when loaded in accordance with 3.9, and on the slopes defined in 3.10.3, 3.10.4, and 3.10.5.

3.10.10 Mobility requirements. The ILJS, when at an 18-inch towing height and loaded in accordance with 3.9, shall be capable of the towing mobility requirements as specified in table I. The test mileage shall be accumulated with a fully loaded ISO container. In addition, the ILJS, when at an 18-inch towing height and loaded in accordance with 3.9, shall be tested for backing up (see 3.10.6) on a level paved surface for a distance of not less than 50 ft, and on a level unpaved surface for a distance of not less than 50 ft. An unpaved surface is defined as an unpaved graded dirt road having a California Bearing Ratio of 40.

TABLE I. Towing mobility requirements.

Surface course	Maximum speed (mph)	Average speed (mph)	Distance (miles)
Paved surfaces	5	2.5	1
Unpaved surface	2	1	0.5

3.11 Towing vehicle (prime mover). The ILJS shall be compatible with all military tactical wheeled vehicles of 5-ton rating or greater with a pintle. Compatibility shall be determined by the ability of the ILJS to meet all the requirements stated herein.

3.12 Maintenance characteristics. The ILJS design, together with components and accessories, shall permit ready access to all items requiring scheduled service or maintenance in the field. Scheduled service and maintenance shall be accomplished with the use of common, non-metric tools. The adjustment of components and accessories shall be accomplished with no disturbance to other components. Grease fittings shall be easily accessible.

3.13 Environmental conditions. The ILJS shall be capable of withstanding the following natural environmental conditions:

a. Ambient temperature conditions - storage, transport, operation, and non-operation shall be -30 degrees to +120 degrees Fahrenheit.
b. Relative humidity - a maximum relative humidity of 100 percent.

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c. Sand and dust - when exposed to sand and dust as specified in MIL-STD-210 for operation in close proximity to aircraft.
d. Salt fog - when exposed to any salt fog atmosphere.

3.14 Transportability. The ILJS shall be capable of being contained on one pallet with dimensions conforming to MIL-P-27443, type I (88 inches X 108 inches, commonly referred to as a 463L pallet) and of being transported without damage on the ramp pallet position on the C-130 and C-141 aircraft. It shall also be capable of being transported by worldwide highway, rail, and ship as specified herein and established by MIL-STD-1366.

3.15 System safety requirements. The ILJS shall have an automatic self-locking failsafe device that will prevent an inadvertent or accidental lowering of the ILJS. The failsafe device must provide positive stop action in the event of a mechanical as well as a procedural accident and should activate within 1/16-inch. Failure modes to be considered must include failure at the point of contact between the lifting mechanism and the container (i.e., rack and spur gear, rack and ratchet pawl, or chain and chain attachment, etc.).

3.16 Human performance and engineering. The ILJS design for operation and maintenance shall provide an environment which fosters effective task performance and personnel safety with minimum training. The ILJS shall be designed such that all operational and scheduled maintenance tasks can be completed by personnel from the 5th to 95th percentile male/female wearing arctic garments as defined in MIL-STD-1472.

3.16.1 Lifting limits. No component of the ILJS shall exceed the three-man design weight limits of 227 lb in accordance with MIL-STD-1472. The ILJS components shall have handles and/or grasp areas which conform to MIL-STD-1472.

3.16.2 Attachment/detachment of ILJS. At no time shall the attachment/detachment of the ILJS from the container and during any operational scenario require an individual to be on top of or beneath the container. The following human engineering requirements shall apply to ISO containers located on the ground or on a flat bed truck or 25K-loader.

3.16.2.1 Attachment of ILJS. Each corner jack assembly shall be attached to the container in not more than ten minutes. The attachment of the ILJS to the container shall not require an individual on top of or beneath the container.

3.16.2.2 Detachment of ILJS. Each jack shall be detached from the container and broken down to the shipping mode with common hand tools in not more than ten minutes.

3.17 Reliability. The ILJS shall have a specified reliability of not less than 0.98 when subjected to the operational cycle specified in 3.10.7. A failure analysis shall be performed in accordance with the test requirements specified in 4.5.4.9.

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3.18 Maintainability. Total active maintenance, including driver/crew checks and services, shall not exceed 15 minutes per operational cycle (see 3.10.7). Preventive maintenance shall not be required at intervals of less than 12 months or 55 cycles, whichever comes first.

3.19 Cleaning, treatment, and painting. ILJS surfaces shall be blast cleaned in accordance with SSPC-SP10, treated and painted as specified herein. Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion products, or any other interfering substances. As soon as practicable, after cleaning, and before any corrosion product or other coating-interfering material can result, the surface shall be prepared or treated to insure the adhesion of the coating system.

3.19.1 Painting. Painting of ILJS surfaces shall consist of at least one coat of primer and at least two finish coats. The primer shall conform to MIL-P-53022 and shall be applied to a clean dry surface as soon as practicable after cleaning and treating. The dry film thickness of the primer shall be not less than 1.5 mils over the entire surface. Paint for the finish coat shall conform to MIL-C-46168, type IV. Color shall be forest green, No. 34083 in acoordance with FED-STD-595. The minimum dry film thickness of each applied coat shall be not less than 1.8 mils over the entire surface. The dry film thickness of the finish coat shall be not less than 3.6 mils over the entire surface. The total dry film thickness of the finish coat shall be not less than 5.1 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

3.20 Lubrication. Lubrication means shall be provided for all parts of the ILJS normally requiring lubrication. Grease lubrication fittings shall conform to SAE J534. When the use of high lubricating pressure will damage grease seals or other parts, fittings with a pressure release shall be used. The ILJS shall be equipped with lubrication data plate conforming to SAE J753. The lubrication data plate shall be installed adjacent to the identification plate.

3.21 Nameplates and identification marking.

3.21.1 Exterior marking. Exterior marking paint shall conform to MIL-C-46168; color shall be black, No. 37038 in accordance with FED-STD-595; paint shall be applied in accordance with MIL-STD-642 and as follows:

a. Safety and instructional markings, such as tire pressure, shall be letters or numbers no larger than 3/4-inch high.
b. Paint type and application date in letters/numbers shall not be more than 3/4-inch high in a location near the identification plate.

3.21.2 Exterior stampings. Identification numbers and assembly markings shall be low-stress stamped on all sides of all sections or assemblies of the ILJS. Characters shall be uniform, gothic, continuous, and as large as possible but no larger than 1/2-inch high.

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3.21.3 Assembly and disassembly plate. A plate shall be provided giving instructions for the assembly and disassembly of the ILJS. The plate shall be attached by screws, bolts, or rivets in a conspicuous place readily accessible during all stated procedures.

3.21.4 ISO container loading plate. A plate shall be provided giving instructions for the coupling of ILJS to the container, container lifting/lowering, and ILJS uncoupling. The plates shall be attached by screws, bolts, or rivets in a conspicuous place readily accessible during all stated procedures.

3.21.5 Identification plates. Identification plates shall be in accordance with MIL-P-15024/6. Any transportation, instruction, safety, warning, caution, and notice plates shall be provided by the manufacturer. Identification plates shall suitable for the life of the equipment. The size of the plates shall not exceed the area of the structure to which it is attached and shall be legible and readily viewable to service personnel standing on level ground.

3.22 Workmanship.

3.22.1 ILJS fabrication. The material used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.22.2 Forgings. Forgings shall be properly shaped and free from fins, cracks, nicks, flaws, seams, and any other injurious defects which might affect the serviceability of the forgings. Tolerances and gages from metal fits shall conform to standards of commercial practice. Finished contact and bearing surfaces shall be true and exact.

3.22.3 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.22.4 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

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3.22.5 Welders and welding.

3.22.5.1 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall provide the contracting officer with certification that the welder assigned to perform welding responsibilities covered by this specification has passed qualification test as prescribed by AWS D1.1, Section 5, for the type of welding operations to be performed and that such qualifications are effective as defined by the code.

3.22.5.2 Welding. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. All welded spatter and projections on mating surfaces where members are in contact for fabrication or assembly shall be ground to a suitable flat surface. Weld penetration shall be such as to provide transference of maximum design stress through the base metal juncture. Fillet welds shall be provided when necessary to reduce stress concentration. Manual and machine welding processes and materials shall be in accordance with AWS D1.1.

3.22.6 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of section 3. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials are inspected for compliance with all the requirements specified herein and in applicable referenced documents.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).
c. Packaging inspection (see 4.8).

4.2.1 First article inspection. The first article inspection shall be performed on one complete ILJS when a first article is required (see 3.2 and 6.2). This inspection shall include the examination of 4.3, the tests of 4.5.4 and 4.5.5, and the inspections of 4.6, 4.7, and 4.8. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.4, the test of 4.5.5, and the inspections of 4.6 and 4.7. This inspection shall be performed on the samples selected in accordance with 4.3.

4.3 Sampling. Sampling and inspection procedures shall be in accordance with MIL-STD-105. The unit of product shall be one complete ILJS. All ILJS offered for delivery at one time shall be considered a lot for the purpose of inspection. The inspection level S-2 and Acceptable Quality Level (AQL) shall be 2.5 percent defective. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for a complete reinspection. Resubmitted lots shall be reinspected using tightened inspection. If the rejected lot was screened, reinspection shall be limited to the defect causing rejection. If the lot was reprocessed, reinspection shall be performed for all defects. Rejected lots shall be separated from new lots, and shall be clearly identified as reinspected lots.

4.3.1 Sampling for examination. Examination shall be based on inspection level S-2 and AQL of 2.5 percent defective.

4.3.2 Sampling for tests. The tests shall be based on inspection level S-2 and AQL of 2.5 percent defective.

4.4 Examination. Each ILJS shall be examined for compliance with the requirements in section 3 of this military specification. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirement shall constitute one defect.

4.5 Tests. The first article shall receive the tests of 4.5.4 and 4.5.5. Samples selected in accordance with 4.3 shall receive the test of 4.5.5. Failure to pass any test shall constitute cause for rejection.

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4.5.1 Test configuration. During the inspections and tests, except in-process inspections, the ILJS shall be fully assembled.

4.5.2 Acceptance testing and in-process inspection. Each production ILJS shall be subjected to complete acceptance testing and in-process inspection by the contractor in accordance with contractor prepared Government approved plans. The Government at its option may elect to witness/participate in the acceptance testing and in-process inspection.

4.5.3 Methods of testing. The examinations and tests specified herein are the minimum required to determine conformance to the requirements delineated in section 3 of this military specification. Additional testing by the contractor may be required to determine conformance to specification requirements. Testing shall be performed with a fully loaded ISO container as specified in 3.9.

4.5.4 First article tests.

4.5.4.1 Corner jack assemblies. To determine conformance with 3.6, the corner jack assemblies shall be tested in accordance with MIL-S-8512. Each corner jack shall withstand a static load of 20,000 lb without showing signs of deterioration or permanent deformation.

4.5.4.2 Towing assembly. To determine conformance with 3.7, the towing assembly shall be demonstrated and inspected to ensure that the lunette eye conforms to MS51336 and that it is capable of attaching to a prime mover (see 3.11) at any height between 10 and 40 inches above the ground.

4.5.4.3 Weight test. To determine conformance with 3.8.2, the ILJS shall be weighed to ensure that the weight requirements are met.

4.5.4.4 Controllability and slope test. To determine compliance with the controllability and slope requirements contained in 3.10.3, 3.10.4, and 3.10.5, the ILJS, when at an 18-inch towing height and loaded in accordance with 3.9, shall be tested and operated in the surface/slope specified.

4.5.4.5 Level loading. To determine conformance with level loading requirements (see 3.10.7.1), the ILJS with a fully loaded ISO container shall be backed up and the container loaded onto a loading ramp of a C-130 and C-141 aircraft. Unless otherwise specified (see 6.2), the contractor shall provide a mock-up version of the ramp loading areas of the C-130 and C-141 aircrafts, which meet all dimensional and load requirements for each type of aircraft.

4.5.4.6 25K-loader loading and unloading test. To determine conformance with 25K-loader loading requirements (see 3.10.7.2), the ILJS with a fully loaded ISO container shall be backed up and the container loaded onto a 25K-loader, and the jacks and hardware removed. To determine conformance with

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25K-loader unloading requirements (see 3.10.7.2), the ILJS shall be attached to a fully loaded ISO container shall be backed up and the container loaded onto a 25K-loader, and the jacks and hardware removed. Contractor shall demonstrate the capability of loading and unloading a fully loaded container (see 3.10.7.2) onto and off of a 25K-loader. Unless otherwise specified (see 6.2), the contractor shall provide a mock-up version of the 25K-loader.

4.5.4.7 Wind load analysis. To determine compliance with 3.10.8, the contractor shall have available to the Government an analysis of the effect a 30 mph wind load on a fully loaded container (see 3.9) while loading/ unloading when the ILJS is at full height extension.

4.5.4.8 Environmental conditions. To determine compliance with 3.13, the ILJS shall be tested in accordance with MIL-STD-810, as specified in table II.

TABLE II. Test methods for environmental conditions.

Specification requirements	MIL-STD-810 Method Test
Ambient temperature	501.2, 502.2, Procedure II
Relative humidity	507.2 Procedure II
Sand and dust	510.2
Salt fog	509.2

4.5.4.9 Load/unload reliability. To determine compliance with 3.10.7, all operational cycles shall be performed twice and a failure analysis of all parts shall be done using the test data to demonstrate one year's usage (i.e., 55 cycles).

4.5.5 Structural integrity of the ILJS. To determine compliance with 3.5, the ILJS with the container jacked to 66 inches above the ground shall be statically tested with a 44,800 lb evenly distributed test payload. The ILJS shall be statically tested with a 25,000 lb eccentric payload located within the cross-hatch area in figure 1 (see 3.9). This latter test shall be conducted three times, with the eccentric payload located in a different position each time.

4.6 In-process inspection. During fabrication of the first production ILJS, in-process inspections shall be conducted to evaluate conformance of materials and workmanship to requirements of specified documents. The ILJS and all components shall be available for inspection. These inspections shall

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be made at the contractor or sub-contractor's facilities prior to application of primer and paint. Processing and welding procedures, quality procedures shall be reviewed during the in-process inspections. Deficiencies reported on this inspection shall be corrected by the contractor prior to completion of the first ILJS production.

4.7 Completed ILJS inspection.

4.7.1 Contractor inspection. The first article ILJS shall be tested and inspected by the contractor. The first article ILJS testing shall include as a minimum the requirements referenced in table III of this military specification. The contractor shall retain the first article unit until delivery of the last production unit. the first article unit shall be available for inspection and comparison with production units by Government inspectors. The contractor may deliver the first article unit to the Government as the final ILJS contract deliverable. If delivered, the first article test unit shall be refurbished to like new condition. The Government at its option may elect to witness and/or participate in contractor testing.

4.7.2 Repair of defects. Defects found as a result of testing shall be corrected by the contractor at no cost to the Government. Failure of the contractor to promptly correct defects shall be cause for suspension of acceptance of production ILJS until corrective action has been accomplished.

4.7.3 Approval and acceptance. Approval and acceptance by the Government of the first article ILJS shall be withheld until the first article testing has been completed and a final determination has been made regarding conformity of the ILJS to requirements including, but not limited to, workmanship and materials.

4.7.4 Mobility requirements. To show compliance with 3.10.10, the ILJS, when at an 18-inch towing height and loaded in accordance with 3.9, shall be tested for towing mobility over distance and courses specified in table I; and shall be tested for reverse (backing up) mobility, as specified in 3.10.10.

4.8 Packaging inspection. The preservation, packing, and marking of the item shall be inspected to verify conformance to the requirements of section 5.

5. PACKING

5.1 Preservation. The levels of preservation shall be level A or commercial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. The ILJS shall be cleaned in accordance with process C-1 and dried in accordance with procedure D-1, D-4, or D-5, as applicable, of MIL-P-116.

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5.1.1.2 Unit protection. Each ILJS shall be preserved in accordance with MIL-P-116, method I. Exterior surfaces subject to corrosion shall be coated with type P-2 preservative. Each unit shall be disassembled when practicable and fully retracted. The towbar shall be secured to the jack leg. Attaching parts accompanying the disassembled unit shall be preserved, bagged, or otherwise secured, appropriately identified, and attached to, or adjacent to the attached fitting.

5.1.2 Commercial. Each ILJS shall be preserved and packaged in a manner which will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity. This level may conform to the supplier's commercial practice when such meets the requirements of this level.

5.2 Packing. Packing shall be level A, B, or commercial, as specified (see 6.2).

5.2.1 Level A. The ILJS jack shall be packed in a wood-cleated box with a load bearing base (skidded) conforming to MIL-B-26195, type II (overseas). The superstructure of the box shall be fabricated in accordance with PPP-B-601. Closure and strapping shall be flat metal strapping in accordance with appendix of PPP-B-601.

5.2.2 Level B. Level B shall be the same as level A, except the container shall conform to MIL-B-26195, type I (domestic).

5.2.3 Commercial. Each ILJS shall be packed in a manner which affords adequate protection, at the lowest rate, against damage during direct shipment from the supply source to the first receiving activity. This level shall conform to applicable carrier rules and regulations and may be the supplier's commercial practice provided such meets the requirements of this level.

5.3 Marking. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. This ILJS is intended for use with ISO containers as a lifting and loading jack system, and mobility and maneuverability of ISO containers.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in acquisition documents:

a. Title, number, and date of this specification.
b. When a first article is required (see 3.2 and 4.2.1).
c. When the contractor shall not provide a mock-up version of aircraft ramp loading areas (see 4.5.4.6).
d. When the contractor shall not provide a mock-up version of the 25K-loader (see 4.5.4.7).
e. Level of preservation and packing (see 5.1 and 5.2).

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6.3 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.475-1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data shall be delivered by the contractor in accordance with the contract or purchase order requirements.

6.4 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.5 Subject term (key word) listing.

ISO container
Aircraft loading jack system
Loading jack system
Lifting jack system

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing Activity:
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(Project 3950-N019)

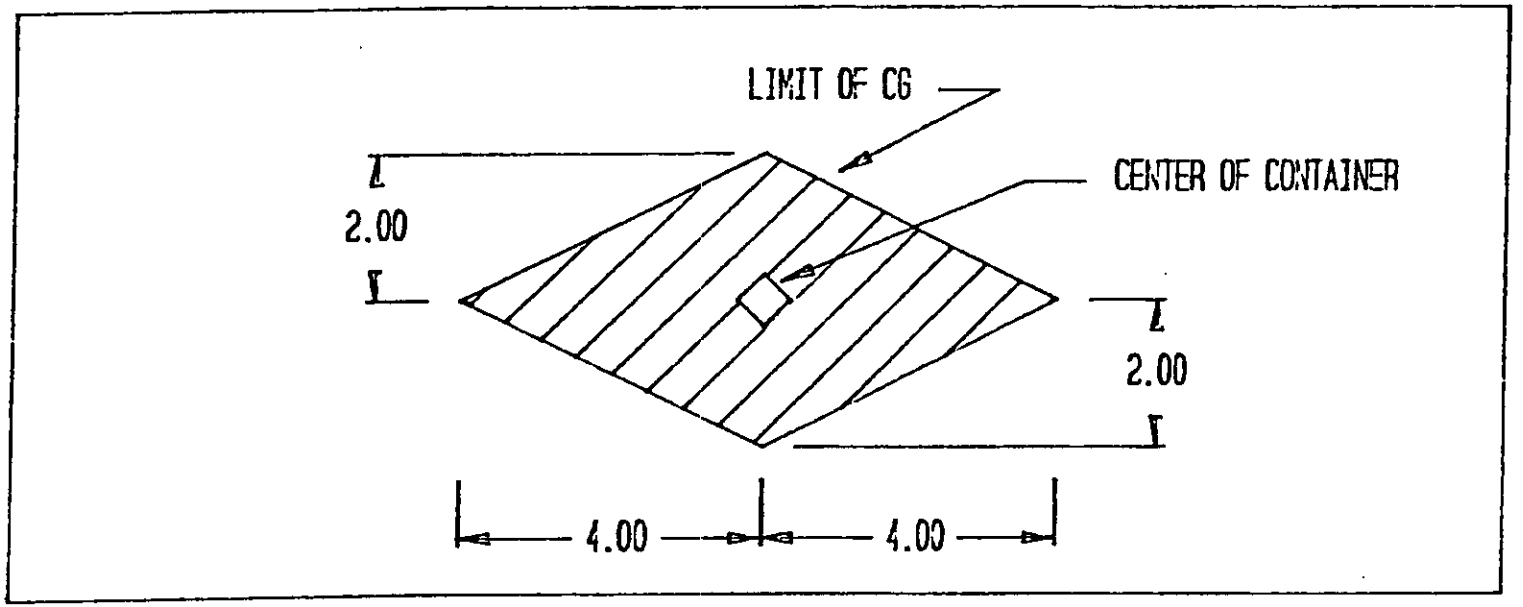
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TABLE III. Completed first ILJS inspection.

Paragraph number/title		I	A	D	T
3.5.1	Attaching points		X	X	
3.6	Corner jack assemblies	X	X	X	4.5.4.1
3.6.3	Tires	X	X		
3.6.4	Brakes		X	X	
3.7	Towing assembly	X	X	X	4.5.4.2
3.8.1	Dimensions and clearance			X	
3.8.2	ILJS weight			X	4.5.4.3
3.10.1	Lifting with ILJS		X	X	
3.10.3, 3.10.4, and 3.10.5	Controllability and slope		X	X	4.5.4.4
3.10.7.1	Level loading		X	X	4.5.4.5
3.10.7.2	25-K loader loading		X	X	4.5.4.6
3.10.8	Wind load		X		4.5.4.7
3.10.9	Parking brake performance		X	X	
3.10.10	Mobility requirements			X	4.7.4
3.11	Towing vehicle			X	
3.12	Maintenance characteristics		X	X	
3.13	Environmental conditions		X	X	4.5.4.8
3.14	Transportability			X	
3.15	System safety requirements		X	X	
3.16	Human factors		X	X	
3.16.1	Lifting limitations		X	X	
3.16.2.1	Attachment of ILJS			X	
3.17	Reliability		X	X	4.5.4.9
3.18	Maintainability		X		
3.19	Painting	X			
3.20	Lubrication	X	X		
3.21	Nameplates and markings	X			
3.22	Workmanship	X			
3.22.1	ILJS fabrication	X	X		

NOTE: I - Inspection; A - Analysis; D- Demonstration; T - Test.

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NOTES: 1. Top view of container.
2. Cross hatch area indicates center of gravity location.
3. All dimensions in feet.

FIGURE 1. Center of gravity.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-L-29250A	LOADING JACK SYSTEM ISO CONTAINER

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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